

ALDEN WU

+1 (408) 832-8816 | contact@aldw.net | aldw.net | github.com/goodtrailer | updated October 9, 2026

EDUCATION

University of California, San Diego

Computer Science M.S. (Depth: Graphics & Vision)

Computer Science B.S. & Mathematics B.S., GPA 3.92

San Diego, CA

Sep 2025 – Jun 2027

Sep 2022 – Jun 2026

EXPERIENCE

UC San Diego – Computer Graphics Researcher (Student) | Python, PyTorch, CUDA, ROCm

Mar 2026 – Sep 2026

- Investigated mechanistic algorithms based on **AI diffusion models** for **2D & 3D content generation**
- Engineered asymptotic performance improvements (observed up to **262,000x speedups** in practice) by adapting efficient 2D image algorithms (e.g. PatchMatch) to 3D voxel domains with GPU acceleration
- Explored efficient representations for 3D data: **implicit surfaces**, **NeRFs**, point clouds, etc.

Amazon – Software Engineer (Intern) | Linux, Android, C++, Python, AWS, LLM, WebDriver

Jun 2025 – Sep 2025

- Worked on **AI/LLM-based automation framework** for Kindle device to facilitate CI/CD
- Increased performance by **7x** for navigation & keypresses by adding new capabilities to the **agentic action model**
- Decreased cost by **5.5x** for common actions by optimizing LLM harness interaction logic
- Improved stability of device-side daemon by fixing **memory buffer management**

Marvell – Software Engineer (Intern) | PHP, HTML/CSS, JavaScript, Subversion, XAMPP

Jun 2024 – Sep 2024

- Refactored and simplified large portions of code to improve maintainability, reducing bloat and repetition
- Used MySQL to display more detailed and useful information to end users

PROJECTS

Path tracer, 3D renderer – aldw.net/projects#pbr, github.com/goodtrailer/lajolla | C++, NVIDIA OptiX

Jan 2025 – Feb 2025

- Physically based Monte-Carlo ray tracer, **GPU accelerated** with NVIDIA OptiX
- Implemented the Smith-GGX **microfacet model** for reflection and transmission
- Improved performance with BSDF importance sampling and next event estimation (MIS)
- Volumetric rendering of **chromatic heterogeneous media** (e.g. colored smoke)

Physics/geometry simulations – aldw.net/projects/#physics | C++, Houdini, NumPy, SciPy

Jan 2024 – Jun 2024

- Derived and implemented the equations of motion for a triple pendulum using **Lagrangian mechanics**
- Developed **numerical solvers**/simulations for **nonlinear ODE** modeling complex multi-body dynamics
- Simulated 2D fluids using vortex methods, applying Kelvin's circulation theorem and **discretizing** the Biot-Savart law
- Computed geodesic distances on **curved geometry** using the heat method (discretizing and integrating the heat flow)

AR/VR application – “AILA” | C#, .NET, Unity, Blender, OpenXR, Meta Quest 2

Apr 2024 – Jun 2024

- Delivered an **intuitive UX** by experimenting with manipulation modes (raycasts, go-go hands, WIM)
- Improved **comfort** by implementing multiple travel methods (teleportation, joystick, skiing) to fit user preference

Strategy game – github.com/goodtrailer/paper | C++, Unreal Engine, HLSL, Blender

Nov 2019 – Feb 2021

- Supported online play with over 3000 pieces and 6 players concurrently using UE4 replication
- Boosted performance by implementing fast 2D algorithms (e.g. A* pathfinding, Bresenham line drawing)

RELEVANT COURSEWORK

CSE Robotics, Embedded Systems (FPGA), Differentiable Programming, Physics Simulation, Computer Vision, Virtual Reality, Computer Graphics, Rendering (PBR), Discrete Differential Geometry (DDG), Animation, Image Processing, Networked Services, Operating Systems, Parallel Programming, Computer Architecture, PL/Compilers, Optimization (ML), Cryptography

MATH Fourier/Harmonic Analysis, Functional Analysis, Linear Algebra, Group Theory, Ring Theory, Galois Theory, Graph Theory, Numerical Analysis, Computational Stochastics, Probability, Vector Calculus, Real Analysis, Algebraic Topology

TECHNICAL SKILLS

Languages C/C++, ARM Assembly, Verilog, MATLAB, C#, Java, Python, JavaScript, HTML/CSS, PHP, PostgreSQL

Frameworks Unreal Engine, Unity, React.js, Express.js, Win32, JUnit, GoogleTest, Appium/Selenium

Developer tools Claude Code, Codex, git, ssh, gdb, AWS, AWS Bedrock, NVIDIA Nsight, CMake, vcpkg, NuGet, Maven, Linux

Libraries/etc. .NET, OpenGL, HLSL, CUDA, ROCm, OpenCL, NVIDIA OptiX, AMD HIP, PyTorch, OpenCV, NumPy, SciPy